Enumeration of Polygon-Free Mosaics

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aren't we using "lattices" otherwise? Let's be consistent.

"Hong and Oh" (unclear — "they" seems to refer to "mosaics" or "lattices")

Abstract

Hong and Oh introduced a model for multiple ring polymers in physics in which an $m \times n$ retangular lattice is constructed from a selection of 7 distinct tiles. These lattices are called *mosaics*. They provided bounds on ~~a subset of~~ these mosaics that both contain polygons and have all other tiles that are not part of a polygon set to the blank tile. We introduce and enumerate mosaics with the related property of not containing a polygon, which we call polygon-free mosaics.

the number of

in which the tile patterns form polygons

Should rephrase and also say we're giving an exact count of what Hong & Oh provided bounds for

1 Mosaics

Imagine you are tasked with tiling a rectangular bathroom floor that is m units by n units, blindfolded. At your disposal is an unending supply of 7 distinct types of tiles. These tiles, diagrammed in Figure 1, are composed of unit squares with dotted lines connecting 2 sides at their midpoints, as well as the "blank" tile T_0.

people will think of uniform probability distributions & calculating probabilities rather than counting

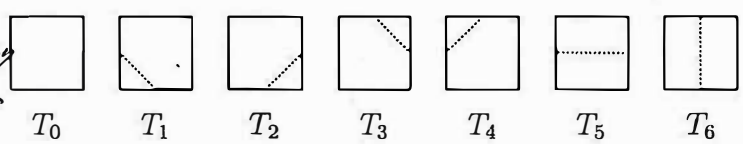


Figure 1: The tile set $\mathbb{T}$

aren't we doing more than this?

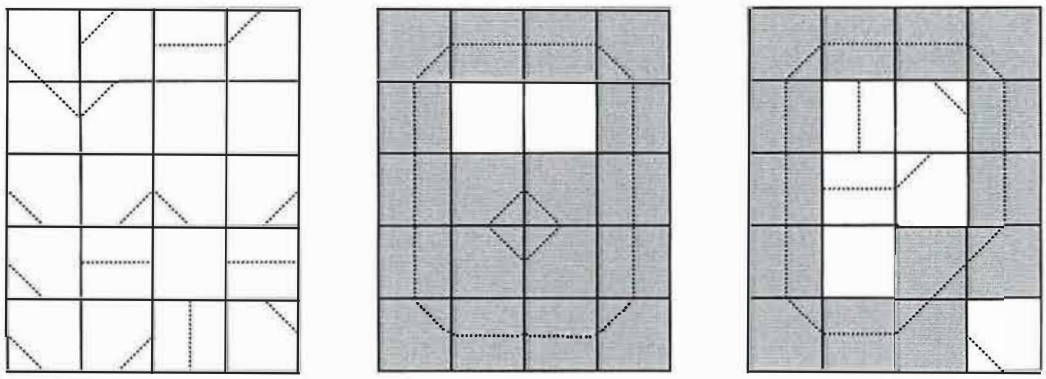
We denote the set of tiles $\mathbb{T} = \{T_0, \ldots, T_6\}$. The task is complete once you place mn ~~randomly selected~~ tiles from $\mathbb{T}$ to cover the floor, after which you remove your blindfold. We are interested in the number of floors that do not contain polygons[1].

We call a fully tiled $m \times n$ floor an (m, n) *mosaic*, where m, n are positive integers. If $m = 5$ and $n = 4$, you may have constructed any of the three mosaics in Figure 2. We highlight the tiles that are part of a polygon in gray.

This sounds like only these three are possible. "Figure 2 shows various (5,4) mosaics." is much simpler.

[1] Polygons are more commonly referred to as "self-avoiding polygons" in the literature to highlight their connection with self-avoiding walks.

add citation



(a) A polygon-free mosaic (b) A polygon mosaic (c) A messy polygon mosaic

Figure 2: Examples of ~~mosaics of size (5, 4)~~ (5, 4) mosaics ~~made of tiles in $\mathbb{T}$~~ redundant

Probably better as one definition. Also: do we still need "messy"? Why?

Definition 1.1. An (m, n) *polygon-free mosaic* is an (m, n) mosaic that does not contain a polygon.

Definition 1.2. An (m, n) *polygon mosaic* is an (m, n) mosaic such that any tile that is not part of a polygon is T_0. Notice this includes the mosaic of all T_0 tiles.

Unless we reference 1.2, probably not a formal example but just text. Also: probably combine with ref to Fig 2 at bottom of page 1

Definition 1.3. An (m, n) *messy polygon mosaic* is an (m, n) mosaic that contains at least one polygon.

Example 1.1. Figure 2a depicts a polygon-free mosaic, Figure 2b depicts a polygon mosaic, and Figure 2c depicts a messy polygon mosaic.

Why define?

~~Therefore~~, let $\mathbb{P}^{(m,n)}$ be the number of (m, n) polygon-free mosaics. Clearly the number of (m, n) messy polygon mosaics is then $7^{mn} - |\mathbb{P}^{(m,n)}|$. To determine the number of polygon-free mosaics, we begin by defining a related object to mosaics, that being binary lattices.

confusion on whether $\mathbb{P}^{(m,n)}$ is a number or a set

2 Binary Lattices

I thought there would be "lattices"

For positive integers m, n, an (m, n) *binary lattice* is a rectangular lattice of $m + 1$ by $n + 1$ vertices, with each vertex labeled 0 or 1. We also define a *framed lattice* to be a binary lattice in which the boundary vertices are labeled 0. An example of a $(5, 7)$ framed lattice is shown on the right of Figure 3.

All text (better?) or all in Definition environment.

Definition 2.1. Let a *cell* be a $(1, 1)$ binary lattice.

For ~~convenience~~, we denote a cell by the 2×2 matrix of its vertex labels. There are ~~sixteen~~ cells: When referring to cells, we omit the connecting lines. There are sixteen cells:

$$\begin{matrix}0&0\\0&0\end{matrix}, \begin{matrix}0&0\\0&1\end{matrix}, \begin{matrix}0&0\\1&0\end{matrix}, \begin{matrix}0&0\\1&1\end{matrix}, \begin{matrix}0&1\\0&0\end{matrix}, \begin{matrix}0&1\\0&1\end{matrix}, \begin{matrix}0&1\\1&0\end{matrix}, \begin{matrix}0&1\\1&1\end{matrix}, \begin{matrix}1&0\\0&0\end{matrix}, \begin{matrix}1&0\\0&1\end{matrix}, \begin{matrix}1&0\\1&0\end{matrix}, \begin{matrix}1&0\\1&1\end{matrix}, \begin{matrix}1&1\\0&0\end{matrix}, \begin{matrix}1&1\\0&1\end{matrix}, \begin{matrix}1&1\\1&0\end{matrix}, \text{ and } \begin{matrix}1&1\\1&1\end{matrix}.$$

not OK!

plural singular

Definition 2.2. Let $\mathcal{L}$ map (m,n) polygon mosaics to a (m,n) binary lattice with the following rule. If the vertex is surrounded by the red dotted lines of an even number of polygons (including 0 polygons), label it 0. If the vertex is surrounded by the red dotted lines of an odd number of polygons, label it 1. Removing the red dotted lines from the tiles gives the framed lattice.

One-direction arrow

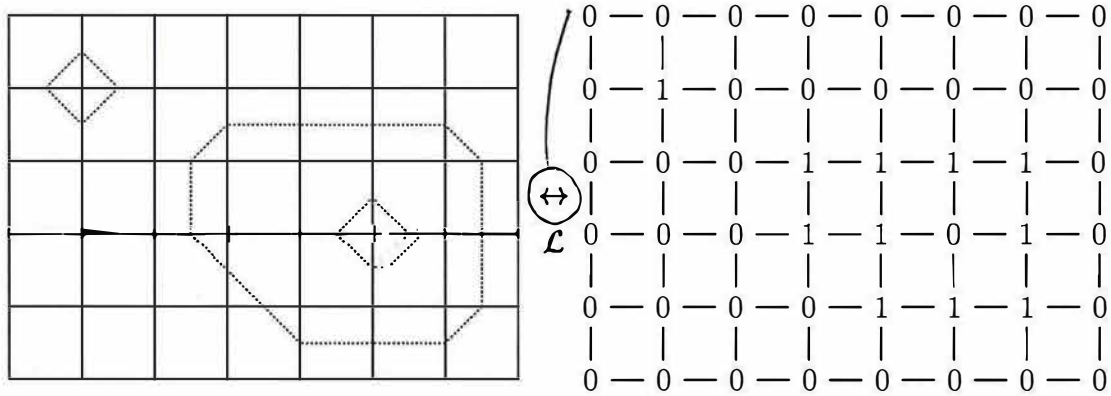


Figure 3: $\mathcal{L}$ applied to a polygon mosaic

Perhaps include in def at beginning of section. In that case we'd observe that the image under $\mathcal{L}$ is a proper lattice.

Notice that under $\mathcal{L}$, no tile can map to the cells $\begin{smallmatrix}0&1\\1&0\end{smallmatrix}$ or $\begin{smallmatrix}1&0\\0&1\end{smallmatrix}$.

Definition 2.3. Let a *proper lattice* be a binary lattice that does not contain any $\begin{smallmatrix}0&1\\1&0\end{smallmatrix}$ or $\begin{smallmatrix}1&0\\0&1\end{smallmatrix}$ cells.

Proposition 2.1. *The map $\mathcal{L}$ is a bijection between (m,n) polygon mosaics and (m,n) proper framed lattices.*

rephrase because there isn't a different $\mathcal{L}$ for each (m,n).

Proof. **TODO** □

"The map $\mathcal{L}$ bijectively maps the set of (m,n) polygon mosaics to the set of (m,n) proper framed lattices for all pos integers m and n."

3 Counting Polygon Mosaics

(too vague)

We now define a ~~general~~ function v that maps a cell to some integer, then extend v to (m,n) binary lattices by taking the product over v of the mn individual cells in the binary lattice.

Delete paragraph as unnecessary.

For polygon mosaics, we use the following definition for v.

Definition 3.1. Define a map v from cells to integers as follows.

$$\begin{cases} 0 & \text{for cells } \begin{smallmatrix}0&1\\1&0\end{smallmatrix}, \begin{smallmatrix}1&0\\0&1\end{smallmatrix} \\ 1 & \text{otherwise} \end{cases}$$

More generally, we define v from binary lattices to integers by taking the product of v applied to each cell in the binary lattice.

This v is not maintaining its identity throughout the paper, so define it in Theorem 1. Further, the definition is needlessly complicated, right?

v is ~~the~~ 1 for a proper lattice and 0 otherwise.

This is pretty trivial and probably should be a proposition. Or better: just a remark. Perhaps best: don't even define v yet!

Theorem 1. *The number of (m,n) polygon mosaics is*

$$\sum v(\ell),$$

where the sum is over all framed lattices ℓ.

Proof. From Proposition 2.1, there is a bijection between polygon mosaics and framed proper lattices, and so $v(\ell)$ counts one poylgon mosaic for each framed proper lattice ℓ. By Definition 3.1, if ℓ' is a binary lattice that contains either $\begin{smallmatrix}0&1\\1&0\end{smallmatrix}$, $\begin{smallmatrix}1&0\\0&1\end{smallmatrix}$, then $v(\ell') = 0$. Therefore, $\sum v(\ell)$ over all framed lattices is equal to the sum over all proper framed lattices, which completes the proof. □

We show that we can more efficiently compute the sum $\sum v(\ell)$ from Theorem 1 using matrices. For matrices, mosaics, and later binary lattices, we number the rows $0, 1, \ldots, m-1$ top to bottom, and columns $0, 1, \ldots, n-1$ left to right.

4 Computing $\sum v(\ell)$ with Matrices

Need to make it clear that this is not just for the above v.

Define $A_1 = \left(v\left(\begin{smallmatrix}0&0\\0&0\end{smallmatrix}\right)\right), B_1 = \left(v\left(\begin{smallmatrix}0&0\\1&0\end{smallmatrix}\right)\right), C_1 = \left(v\left(\begin{smallmatrix}1&0\\0&0\end{smallmatrix}\right)\right), D_1 = \left(v\left(\begin{smallmatrix}1&0\\1&0\end{smallmatrix}\right)\right)$, and for integers $k \geq 1$,

Say this more clearly, and above

how about changing k to n to match proof?

$$A_{k+1} = \begin{pmatrix} v\left(\begin{smallmatrix}0&0\\0&0\end{smallmatrix}\right)A_k & v\left(\begin{smallmatrix}0&0\\0&1\end{smallmatrix}\right)B_k \\ v\left(\begin{smallmatrix}0&1\\0&0\end{smallmatrix}\right)C_k & v\left(\begin{smallmatrix}0&1\\0&1\end{smallmatrix}\right)D_k \end{pmatrix} \quad B_{k+1} = \begin{pmatrix} v\left(\begin{smallmatrix}0&0\\1&0\end{smallmatrix}\right)A_k & v\left(\begin{smallmatrix}0&0\\1&1\end{smallmatrix}\right)B_k \\ v\left(\begin{smallmatrix}0&1\\1&0\end{smallmatrix}\right)C_k & v\left(\begin{smallmatrix}0&1\\1&1\end{smallmatrix}\right)D_k \end{pmatrix}$$

$$C_{k+1} = \begin{pmatrix} v\left(\begin{smallmatrix}1&0\\0&0\end{smallmatrix}\right)A_k & v\left(\begin{smallmatrix}1&0\\0&1\end{smallmatrix}\right)B_k \\ v\left(\begin{smallmatrix}1&1\\0&0\end{smallmatrix}\right)C_k & v\left(\begin{smallmatrix}1&1\\0&1\end{smallmatrix}\right)D_k \end{pmatrix} \quad D_{k+1} = \begin{pmatrix} v\left(\begin{smallmatrix}1&0\\1&0\end{smallmatrix}\right)A_k & v\left(\begin{smallmatrix}1&0\\1&1\end{smallmatrix}\right)B_k \\ v\left(\begin{smallmatrix}1&1\\1&0\end{smallmatrix}\right)C_k & v\left(\begin{smallmatrix}1&1\\1&1\end{smallmatrix}\right)D_k \end{pmatrix}.$$

We work with general v in this section, and substitute a specific v later.

Definition 4.1. Let the n digit binary representation of the ~~number k~~ nonnegative integer $k < 2^n$ be written as $\beta_n(k)$, with $\beta_0(0)$ being the empty string. ~~If n is 0, $\beta_n(k)$ returns the empty string~~.

maybe we should say this all together at the beginning

We extend our shorthand for cells to $(1, n)$ binary lattice using analagous matrix notation and the definition for $\beta_n(k)$. For example, we write

$$v\begin{pmatrix} 0 & \beta_2(1) & 0 \\ 0 & \beta_2(0) & 0 \end{pmatrix} = v\begin{pmatrix} 0&0&1&0 \\ 0&0&0&0 \end{pmatrix} = v\begin{pmatrix} 0&0 \\ 0&0 \end{pmatrix} v\begin{pmatrix} 0&1 \\ 0&0 \end{pmatrix} v\begin{pmatrix} 1&0 \\ 0&0 \end{pmatrix}.$$

Proposition 4.1. *For all integers $n \geq 1$, the (i,j)-th entry of*

$$A_n \text{ is } v\begin{pmatrix} 0 & \beta_{n-1}(i) & 0 \\ 0 & \beta_{n-1}(j) & 0 \end{pmatrix}, \quad B_n \text{ is } v\begin{pmatrix} 1 & \beta_{n-1}(i) & 0 \\ 0 & \beta_{n-1}(j) & 0 \end{pmatrix},$$

$$C_n \text{ is } v\begin{pmatrix} 0 & \beta_{n-1}(i) & 0 \\ 1 & \beta_{n-1}(j) & 0 \end{pmatrix}, \quad D_n \text{ is } v\begin{pmatrix} 1 & \beta_{n-1}(i) & 0 \\ 1 & \beta_{n-1}(j) & 0 \end{pmatrix}.$$

Difficult to read. Can it just be all inline text? Otherwise "(i,j) entry" gets lost.

Let i,j be nonnegative integers less than 2^{n-1}.

Proof. We prove by induction on n. The $n = 1$ case is trivial, as $\beta_0(0)$ is the empty string. We next assume our result for some fixed $n \geq 1$. Since A_n has size $2^{n-1} \times 2^{n-1}$, the $(i, j + 2^{n-1})$ entry ~~element~~ of A_{n+1} equals $v\left(\begin{smallmatrix}0&0\\0&1\end{smallmatrix}\right)$ times the (i,j) element of B_n ~~by~~ SITET our induction hypothesis, this equals

① Before prop, explain entries (i,j) where we index starting at 0.
② Statement of prop: "... for all nonnegative integers i,j < 2^{n-1}"

$$v\begin{pmatrix}0&0\\0&1\end{pmatrix}v\begin{pmatrix}0&\beta_{n-1}(i)&0\\1&\beta_{n-1}(j)&0\end{pmatrix}=v\begin{pmatrix}0&0&\beta_{n-1}(i)&0\\0&1&\beta_{n-1}(j)&0\end{pmatrix}=v\begin{pmatrix}0&\beta_n(i)&0\\0&\beta_n(j+2^{n-1})&0\end{pmatrix},$$

as desired. Analagous arguments show that the other blocks in A_{n+1} have the desired values. Similarly, since C_n has size $2^{n-1}\times 2^{n-1}$, the $(i+2^{n-1},j)$ element of D_{n+1} equals $v\left(\begin{smallmatrix}1&1\\1&0\end{smallmatrix}\right)$ times the (i,j) element of C_n. By our induction hypothesis, this equals

$$v\begin{pmatrix}1&1\\1&0\end{pmatrix}v\begin{pmatrix}1&\beta_{n-1}(i)&0\\0&\beta_{n-1}(j)&0\end{pmatrix}=v\begin{pmatrix}1&1&\beta_{n-1}(i)&0\\1&0&\beta_{n-1}(j)&0\end{pmatrix}=v\begin{pmatrix}1&\beta_n(i+2^{n-1})&0\\1&\beta_n(j)&0\end{pmatrix},$$

again as desired. These arguments must be adapted to prove the desired results for all blocks of B_{n+1}, C_{n+1}, and D_{n+1}, but all the work is similar. □

STET

Theorem 4.2. *For all positive integers m and n, the $(0,0)$ entry of A_n^m equals $\sum v(\ell)$, where the sum is over all (m,n) framed lattices ℓ.*

Proof. Fix n. We first introduce some terminology. For all positive integers m, we say a *height m strip* is an (m,n) binary lattice in which all vertices in the outside columns (i.e., column 0 and column n) are labeled zero. Then for a height m strip ℓ and a nonnegative integer $i<2^{n-1}$, we say a particular row of ℓ is *determined by i* if that row is equal to $0\ \beta_{n-1}(i)\ 0$. Note that an (m,n) framed lattice is exactly a height m strip in which both the top and bottom rows are determined by 0. Thus, to prove our proposition it suffices to show the following.

Indent or don't say "Claim" in this way.

Claim. For all positive integers m and nonnegative integers $i,j<2^{n-1}$, the (i,j) entry of A_n^m equals $\sum v(\ell)$, where the sum is over all height m strips ℓ whose first row is determined by i and whose last row is determined by j.

To prove the claim, first observe that for each i and j, there exists a unique height 1 strip whose first row is determined by i and whose last row is determined by j. By Proposition 4.1, v applied to this strip is equal to the (i,j) entry of A_n. This proves our claim when $m=1$. We continue by induction, so for some $m\geq 2$, assume the claim for $m-1$.

non-binary spacent all such places.

Fix nonnegative integers $i,j<2^{n-1}$, and consider some nonnegative integer $k<2^{n-1}$. As in the $m=1$ case, Proposition 4.1 tells us that the (i,k) entry of A_n equals v applied to the unique height 1 strip with first and last rows determined respectively by i and by k. By our induction hypothesis, the (k,j) entry of A_n^{m-1} equals $\sum v(\ell)$, where the sum is over all height $m-1$ strips ℓ whose first and last rows are determined respectively by k and by j. Consider applying v both to the height 1 strip indicated above and to one of the height $m-1$ strips ℓ in this sum. As in Figure (picture below), by the definition of v, the product of these two numbers is equal to v applied to the height m strip formed by overlapping the bottom row of the height 1 strip to the top row of ℓ.

number the figure

vertical gaps should go

$$v\begin{pmatrix}0&\beta_{n-1}(i)&0\\0&\beta_{n-1}(k)&0\end{pmatrix}v\begin{pmatrix}0&\beta_{n-1}(k)&0\\\vdots&\vdots&\vdots\\0&\beta_{n-1}(j)&0\end{pmatrix}=v\begin{pmatrix}0&\beta_{n-1}(i)&0\\0&\beta_{n-1}(k)&0\\\vdots&\vdots&\vdots\\0&\beta_{n-1}(j)&0\end{pmatrix}$$

From this we see that the product of the (i,k) entry of A_n and the (k,j) entry of A_n^{m-1} is equal to $\sum v(\ell)$, where the sum is now over all height m strips ℓ whose first, second, and last rows are determined respectively by i, by k, and by j. This tells us, in turn, that when we additionally sum over all nonnegative $k<2^{n-1}$, we can conclude that the (i,j) entry of

This is a big result and seems hidden as a footnote here. It's achieving what Hong and Oh didn't do, right? Incidentally, can we recreate Hong & Oh's estimate based on our work? I think the real result should look like Th. 5.4 below, not Th. 1 above.

redundant with what is above

$A_n \cdot A_n^{m-1} = A_n^m$ is, as desired, the sum $\sum v(\ell)$ where the sum is over all height m strips ℓ whose first and last rows are determined respectively by i and by j.

Finally, ~~as strips~~ whose first and last rows are determined by $i = j = 0$ are just framed lattices, we get the result. □

Substituting the values for v from Definition 3.1 into the definitions of matrices A_k, B_k, C_k, D_k gives the sum from Theorem 1 by Theorem 4.2. Now that we have enumerated polygon mosaics, we move to the more complicated polygon-free mosaics.

5 Polygon-Free Mosaics

We now redefine v to enumerate polygon-free mosaics.

Definition 5.1. Define a map v from cells to integers as follows.

$$\begin{cases} 7 & \text{for cells } \begin{smallmatrix}0&0\\0&0\end{smallmatrix}, \begin{smallmatrix}1&1\\1&1\end{smallmatrix} \\ 0 & \text{for cells } \begin{smallmatrix}0&1\\1&0\end{smallmatrix}, \begin{smallmatrix}1&0\\0&1\end{smallmatrix} \\ -1 & \text{for cells } \begin{smallmatrix}0&0\\1&0\end{smallmatrix}, \begin{smallmatrix}1&0\\1&1\end{smallmatrix} \\ 1 & \text{otherwise} \end{cases}$$

More generally, we define v from binary lattices to integers by taking the product of v applied to each cell in the binary lattice.

Probably best just in the text and unnumbered.

I count 12 I count 1 -7^{13}

Example 5.1. If we let ℓ be the framed lattice on the right of Figure 3, we have $v(\ell) = -7^{11}$, as there are 2 $\begin{smallmatrix}0&0\\1&0\end{smallmatrix}$ cells, 1 $\begin{smallmatrix}1&1\\0&1\end{smallmatrix}$ cells, 9 $\begin{smallmatrix}0&0\\0&0\end{smallmatrix}$ cells, and 2 $\begin{smallmatrix}1&1\\1&1\end{smallmatrix}$ cells.

We show our choice of v has useful properties by first considering the magnitude, then the sign. For the magnitude of v, for fixed m, n give an index $1 \le i \le N$ to every way one polygon can be formed in an (m, n) mosaic. ~~We remind the reader that~~ A polygon is determined not only by its shape and orientation, but also its location within the mosaic. For example, Figure 4 depicts three distinct polygons in a $(2, 4)$ mosaic, which for reference we call polygons 1, 2, and 3.

It's not a reminder if we have not said it before. Have we?

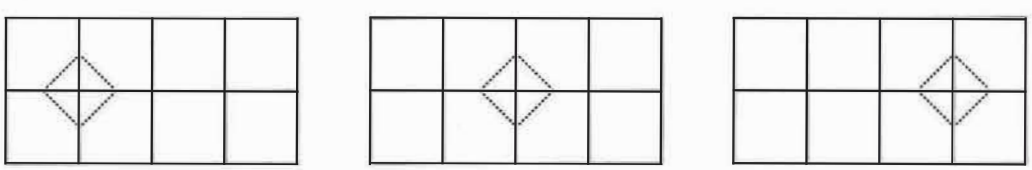


Figure 4: Polygons 1, 2, and 3

Definition 5.2. Let M_i be the set of (m, n) mosaics that contain polygon i.

We immediately have the number of messy polygon mosaics is

$$7^{mn} - \left|\mathbb{P}^{(m,n)}\right| = \left|\bigcup_{i=1}^{N} M_i\right|. \tag{1}$$

Notice that not every subset of polygons $1 \leq i \leq N$ has a mosaic that contains them all. For example, for the polygons in Figure 4 ~~the second from left column contains different tiles in polygon 1 and polygon 2, so no mosaic exists that contain these two polygons.~~ Consequently ~~$M_1 \cap M_2 = \emptyset$.~~ → $M_1 \cap M_2 = M_2 \cap M_3 = \emptyset$ but $M_1 \cap M_3$ contains exactly the polygon mosaic with polygons 1 and 3.

Proposition 5.1. *Let J be a subset of indices of polygons. If there exists at least one mosaic that contains all polygons in J, we have*

$$\left|\bigcap_{j \in J} M_j\right| = \left|v(\ell)\right|, \tag{2}$$

where ℓ is the framed lattice such that $\ell = \cancel{f(M)}$ $\mathcal{L}(M)$ where M is ~~any~~ the mosaic that only contains the polygons in J.

I think an example would be much better. Possibly in addition to this, but probably better in place of this.

Proof. From the definition of M_j, the left hand side of Equation 2 counts the number of mosaics that necessarily contain the polygons in J, and possibly contain other polygons. For the right hand side, if a cell comes from a tile that is part of a polygon, the absolute value of v of the cell is 1. Therefore, $|v(\ell)|$ is just 7 raised to the number of $\begin{smallmatrix}0&0\\0&0\end{smallmatrix}$ and $\begin{smallmatrix}1&1\\1&1\end{smallmatrix}$ cells in ℓ. As all tiles in $\mathbb{T}$ can map to $\begin{smallmatrix}0&0\\0&0\end{smallmatrix}$ and $\begin{smallmatrix}1&1\\1&1\end{smallmatrix}$, $|v(\ell)|$ counts all possible tile assignments in all locations that are not part of a polygon in J. Therefore $|v(\ell)|$ counts mosaics that contain the polygons in J. This is the definition of $\bigcap_{i \in S} M_i$, and so the equality holds. □

that are not part of polygons

Definition 5.3. If ℓ is a framed proper lattice, let ~~$P(\ell)$~~ to be the number of polygons in the polygon mosaic formed by replacing all $\begin{smallmatrix}0&0\\0&0\end{smallmatrix}$ and $\begin{smallmatrix}1&1\\1&1\end{smallmatrix}$ cells with the blank tile T_0, and replacing all other cells with the unique tile that maps to that cell under f.

Not needed. We can just say $|\mathcal{L}^{-1}(\ell)|$ as necessary.

Rephrase based on current statement of 5.2 (i.e., start with P)

For the sign of v, we show that ~~though~~ although $v(\ell)$ is computed "cell-by-cell", we recover global information about $P(\ell)$.

Proposition 5.2. *For a polygon mosaic P, let $\ell = \mathcal{L}(P)$. If P contains an even [resp. odd] number of polygons then $v(\ell)$ is positive [resp. negative].*

"flipping" now seems better

Proof. Let N be the number of vertices of ℓ that are 1, and define ℓ_0 to be the lattice formed by changing these N vertices from 1 to 0. That is, ℓ_0 has the same dimensions as ℓ but all vertices are 0. We will change ℓ_0 back to ℓ by ~~changing~~ these vertices back from 0 to 1, one vertex at a time. The order in which ~~we make these changes~~ is as follows. "flip vertices"

flip

We ~~change back the~~ vertices one row at a time, working from top to bottom. In each row we make two passes, each time moving from left to right, and each time ~~only changing vertices~~ flipping a vertex if the corresponding vertex of ℓ is 1. On the first pass we ~~change a vertex~~ flip if the vertex is directly above it is 1. (Because we are working the rows from top to bottom, this vertex is 1 in both our current lattice and also in ℓ.) On the second pass we ~~change the~~ flip the remaining vertices of the row, i.e., if the vertex is directly above it is 0.

This procedure creates a sequence $\ell_0, \ell_1, \ldots, \ell_N$ of lattices with $\ell_N = \ell$. Because $\ell = \mathcal{L}(P)$, we know ℓ is a proper framed lattice. Since every ℓ_i has the same dimensions as ℓ and has vertices of 0 in at least the same positions as in ℓ, we see every ℓ_i is a framed lattice. We will prove that every ℓ_i is also a proper lattice (which is the reason we make two passes through each row) and that by taking $P_i = \mathcal{L}^{-1}(\ell_i)$, the second part of the proposition holds for P_i and ℓ_i.

You decided to make two propositions from my one.

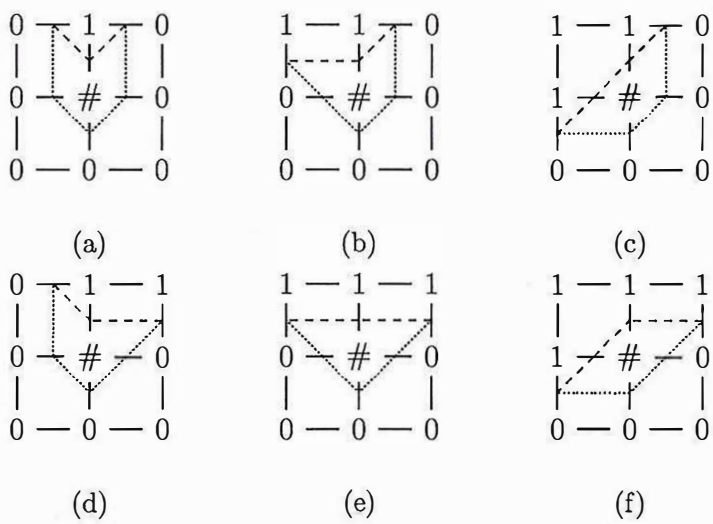
$$\begin{array}{ccccc} ul & - & u & - & ur \\ | & & | & & | \\ l & - & \# & - & r \\ | & & | & & | \\ dl & - & d & - & dr \end{array}$$

Figure 5: Cases Diagram

Then

Because every ℓ_i is framed, in the transition from ℓ_{i-1} to ℓ_i, the single vertex that changes is adjacent to eight vertices, as in Figure 5, where the changing vertex # is 0 in ℓ_{i-1} and 1 in ℓ_i, and the remaining vertices are labeled for their relative position to this central vertex, such as dl for down left and u for up.

We need to consider all possible adjacent vertices. Since our process goes from top to bottom one row at a time, we have $(dl, d, dr) = (0, 0, 0)$. Also, because of the way in which we make our two passes, if $u = 1$ then $r = 0$ and $(ul, l) \neq (0, 1)$, whereas if $u = 0$ then $(ur, r) \neq (0, 1)$. Additionally, suppose $u = 0$, ~~and that~~ neither (ul, l) nor (ur, r) equals $(1, 0)$. Indeed, if $(ul, l) = (1, 0)$ then since we are making the second pass through the row, we would have a cell $\begin{smallmatrix} 1 & 0 \\ 0 & 1 \end{smallmatrix}$ in ℓ_i and thus also in $\ell_N = \ell$, contradicting that ℓ is a proper lattice. Similarly, $(ur, r) = (1, 0)$ implies the that ℓ has a cell $\begin{smallmatrix} 0 & 1 \\ 1 & 0 \end{smallmatrix}$ (since with $ur = 1$, r cannot change in the second pass), again contradicting that ℓ is a proper lattice. Putting all this together, on the first pass, i.e., when $u = 1$, the vertices can only be as in the six cases pictured in Figure 6. Similarly, on the second pass we have $u = 0$ and one of the six cases of Figure 7.



These look great! Also for Figure 7.

Figure 6: ~~Step 2 Cases~~ when $u = 1$

Since trivially ℓ_0 is a proper lattice, the twelve cases in Figures 6 and 7 show inductively that every ℓ_i is a proper lattice. For every i, we have already seen that ℓ_i is a framed lattice, so let $P_i = \mathcal{L}^{-1}(\ell_i)$. Figures 6 and 7 also show the portions of P_{i-1} and P_i that are in these two cells, where the dashed lines are from the polygons of P_{i-1} and the dotted lines from the polygons of P_i. (Recall that # is 0 in ℓ_{i-1} and 1 in ℓ_i.)

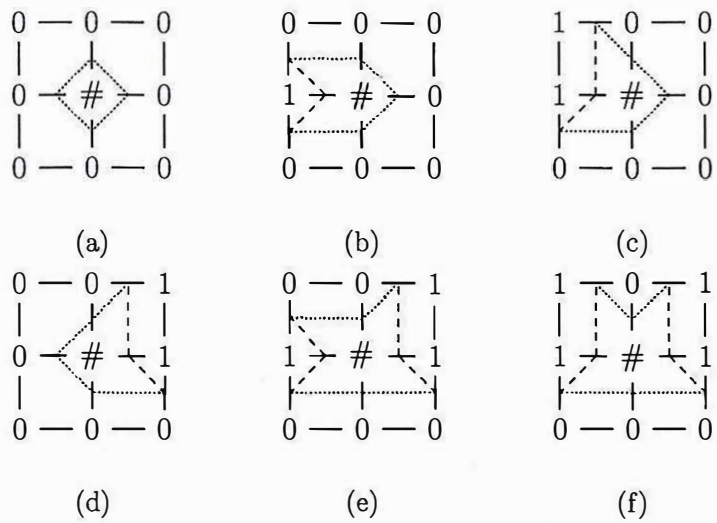


Figure 7: ~~Step 3 Cases~~ when $u = 0$

see above note

Let us say a cell is *negative* if it is $\begin{smallmatrix}0&0\\1&0\end{smallmatrix}$ or $\begin{smallmatrix}1&0\\1&1\end{smallmatrix}$, i.e., if its evaluation under v is negative. The second conclusion of the proposition can be rephrased by saying that the number of negative cells in ℓ and the number of polygons in P have the same parity, i.e., are both even or both odd. For example, ℓ_0 satisfies the conclusion of the proposition because ℓ_0 has no negative cells and P_0 has no polygons. (Second conclusion?) satisfy

Assume now that some P_{i-1} and ℓ_{i-1} ~~satisfies~~ the second conclusion of the proposition, and let us show that the same is true for P_i and ℓ_i. This will complete the proposition. We know that we are in one of the twelve cases of Figures 6 and 7, and we must look at the number of negative cells in ℓ_i versus ℓ_{i-1}, and the number of polygons in P_i versus P_{i-1}. An easy case is (a) of Figure 7 because ℓ_i has one more negative cell (specifically, a cell $\begin{smallmatrix}0&0\\1&0\end{smallmatrix}$), and P_i has one more polygon than P_{i-1}.

Let us now look in the remaining cases at how the number of negative cells changes from ℓ_{i-1} to ℓ_i. In the six cases of Figure 6 and also case (d) of Figure 7, the four pictured cells have no negative cells regardless of whether # equals 0 or 1, so there is no change. The number is also unchanged in (b) of Figure 7 because in this case a single negative cell $\begin{smallmatrix}0&0\\1&0\end{smallmatrix}$ changes position between ℓ_{i-1} and ℓ_i. In (c) of Figure 7, the number of negative cells increases by 2 because changing 0 to 1 in the central position creates both a cell $\begin{smallmatrix}1&0\\1&1\end{smallmatrix}$ and a cell $\begin{smallmatrix}0&0\\1&0\end{smallmatrix}$. For the final two cases, the number decreases by 1 in (e) of Figure 7 and increases by 1 in (f) of Figure 7. Therefore, we can complete the proposition by showing that in the six cases of Figure 6 and in (b), (c), (d) of Figure 7, P_{i-1} and P_i have the same number of polygons, whereas in (e) and (f) of Figure 7, the number of polygons in P_{i-1} and P_i differ by 1.

Consider, for example, (c) of Figure 6, and look at the point just above the lower left corner and the point just left of the upper right corner. In P_{i-1} these are connected by a straight dashed line, and in P_i these are connected by dotted line segments. We call both of these *inside connections.* Because P_{i-1} and P_i are polygon mosaics, these same points are connected by straight line segments outside the pictured area, and we will call these

outside connections because they connect points on the edge of the pictured square with a path outside the square. This outside connection is the same in P_{i-1} and P_i. Therefore P_{i-1} and P_i have the same number of polygons because the polygon with this outside connection changes shape, and all other polygons are unchanged. Similar reasoning completes the remaining cases of Figure 6 and cases (b), (c), (d) of Figure 7.

We now must show that in (e) and (f) of Figure 7, the number of polygons in P_{i-1} and P_i differ by 1. We will explicitly show this in (e), and essentially identical reasoning applies to (f). There are four places where inside connections meet the edge of the outer square. Beginning in the lower left and working clockwise, let x_1, x_2, x_3, and x_4 be these points. For example, x_3 is just left of the upper right corner and x_4 is just above the lower right corner. In P_{i-1}, there is an inside connection between x_1 and x_2, and another between x_3 and x_4, whereas P_i has inside connections between x_1 and x_4, and between x_2 and x_3.

Any outside connections in P_{i-1} and P_i must be identical, and since these are polygon mosaics, each of x_1, x_2, x_3, and x_4 is the end of some outside connection. One possibility is that we have one outside connection between x_1 and x_2, and another between x_3 and x_4. In this case, the inside and outside connections of P_{i-1} complete two polygons, one including x_1 and x_2, and the other including x_3 and x_4. However, P_i will have one fewer polygon because a single polygon contains all x_i. Another possibility is to have one outside connection between x_1 and x_4, and another between x_2 and x_3. In this case, in P_{i-1} a single polygon includes all x_i, whereas in P_i they complete two polygons, one containing x_1 and x_4, and the other containing x_2 and x_3, and therefore P_i has one more polygon than P_{i-1}. These are the only possibilities since it is impossible to have an outside connection between x_1 and x_3, and another between x_2 and x_4. This is because our polygons are in a plane, and polygons in a mosaic are not able to cross one another. This completes our proof. □

Theorem 5.3. *The number of* (m, n) ~~*mosaics that do not contain a polygon has*~~ polygon-free mosaics is

$$\left|\mathbb{P}^{(m,n)}\right| = \sum_{\ell} v(\ell), \tag{3}$$

where the sum is over all (m, n) *framed lattices.*

Proof. We begin by rewriting Equation 1 to

$$\left|\mathbb{P}^{(m,n)}\right| = 7^{mn} - \left|\bigcup_{i=1}^{N} M_i\right|.$$

By the inclusion-exclusion principle (~~proved in~~ see the Appendix), we have

not what you mean — $J \subseteq \mathbb{Z}$

$$-\left|\bigcup_{i=1}^{N} M_i\right| = \sum_{\emptyset \neq J \subseteq \{1,2,\ldots,N\}} (-1)^{|J|} \left|\bigcap_{j \in J} M_j\right|. \tag{4}$$

Consider an individual subset of polygon indices J in the sum in Equation 4 such that $|J| = k \geq 1$. If there exists no mosaic that contains all polygons in J, by the definition of M the term is 0. If at least one mosaic does exist, Propositions 5.1 and 5.2 give ~~the following~~

$$(-1)^{|J|} \left|\bigcap_{j \in J} M_j\right| = v(\ell),$$

"For some $\emptyset \neq J \subseteq \{1, 2, \ldots, N\}$ if $\bigcap_{j \in J} M_j$ is nonempty then ..."

I think separating out by k just makes things needlessly harder.

$\ell = \mathcal{L}(\mathcal{M})$, where $\mathcal{M}$ is the mosaic containing exactly (!!) the polygons M_j, $j \in J$.

where ~~ℓ is the binary lattice such that $f(\mathcal{M}) = \ell$ for all mosaics $\mathcal{M}$ that contain at least the polygons in J.~~

Now discard all 0 terms in Equation 4 and group the terms in Equation 4 for all J such that $|J| = k$. From the definition of M, the absolute value of this collection of terms counts the number of mosaics that have at least k polygons, and so

$$\sum_{J \subseteq \{1,2,\ldots,N\}, |J|=k} (-1)^{|J|} \left| \bigcap_{j \in J} M_j \right| = \sum_{\{\ell | P(\ell)=k\}} v(\ell).$$

I think this can be simplified

Additionally notice that the binary lattice ℓ^* made up of all $\begin{smallmatrix} 0 & 0 \\ 0 & 0 \end{smallmatrix}$ cells has $v(\ell^*) = 7^{mn}$, and is the only framed lattice such that $P(\ell) = 0$. Therefore, we can write

$$\left| \mathbb{P}^{(m,n)} \right| = 7^{mn} - \left| \bigcup_{i=1}^{N} M_i \right| = \sum_{k \geq 0} \sum_{\{\ell | P(\ell)=k\}} v(\ell).$$

Finally, notice that the sum $\sum_\ell v(\ell) = 0$ for the framed lattices that contains $\begin{smallmatrix} 0 & 1 \\ 1 & 0 \end{smallmatrix}$ or $\begin{smallmatrix} 1 & 0 \\ 0 & 1 \end{smallmatrix}$, as $v\left(\begin{smallmatrix} 0 & 1 \\ 1 & 0 \end{smallmatrix}\right) = v\left(\begin{smallmatrix} 1 & 0 \\ 0 & 1 \end{smallmatrix}\right) = 0$. Therefore, summing over all framed lattices in Equation 3 gives the desired result. □

As in Section 4, substituting the values of v into matrices A_k, B_k, C_k, D_k gives the sum from Theorem 5.3 by Theorem 4.2. For completeness, we write these substitutions below.

Definition 5.4. For integers $k \geq 1$ let A_k, B_k, C_k, D_k be $2^{k-1} \times 2^{k-1}$ matrices with integer entries, where $A_1 = (7)$, $B_1 = (-1)$, $C_1 = (1)$, $D_1 = (1)$ and

$$A_{k+1} = \begin{pmatrix} 7A_k & B_k \\ C_k & D_k \end{pmatrix} \quad B_{k+1} = \begin{pmatrix} -A_k & B_k \\ 0C_k & D_k \end{pmatrix}$$
$$C_{k+1} = \begin{pmatrix} A_k & 0B_k \\ C_k & D_k \end{pmatrix} \quad D_{k+1} = \begin{pmatrix} A_k & -B_k \\ C_k & 7D_k \end{pmatrix}.$$

not a def but just part of the statement of Theorem 5.4.

Theorem 5.4. *The number of (m,n) polygon-free mosaics $\left|\mathbb{P}^{(m,n)}\right|$ is the $(0,0)$ entry of A_n^m.*

Finally, we introduce possible extensions to our work, as well as how our work relates to other mosaic enumeration efforts.

6 Related Work and Extensions

In terms of related work, our work can be seen as an extension to Hong and Oh [2] and further generalizing the techniques in Oh et al. [9] to explore a new direction in mosaic enumeration. To begin, Hong and Oh [2] provided bounds on the number of polygon mosaics.

Theorem 6.1 ([2]). *The number of (m,n) polygon mosaics for $m, n \geq 3$ is bounded between $2^{m+n-3}\left(\frac{17}{10}\right)^{(m-2)(n-2)}$ and $2^{m+n-3}\left(\frac{31}{16}\right)^{(m-2)(n-2)}$.*

Does our work imply this? We should discuss.

I don't think this is what you mean. How do Oh know about lattices? Also, aren't their matrices half the size?

Theorem 1 nicely compliments this result with a procedure to exactly enumerate polygon mosaics, as well as exactly enumerating the sequence A181245 on the OEIS [4]. It is important to note that contrary to this work, the authors in [2] did not consider the mosaic of all T_0 a polygon mosaic, and so our enumeration differs by 1. We also point out that Theorem 1 is likely known to the authors of [2] and [9], though to our knowledge it has not appeared in print. nbsp

In related work, Lomonaco and Kauffman [3] introduced mosaics constructed from a tile set of 11 distinct tiles, of which $\mathbb{T}$ is a subset. The authors were interested in a subset of mosaics which they call *knot mosaics*. Oh et al. [9] enumerated the number of knot mosaics using an analagous method to that of Section 4.

Theorem 6.2 ([9]). *The number of (m,n) knot mosaics for $m,n \geq 2$ is $2\,\|(X_{m-2}+O_{m-2})^{n-2}\|$, where $X_0 = O_0 = (1)$ and X_{m-2} and O_{m-2} are $2^{m-2} \times 2^{m-2}$ matrices defined as*

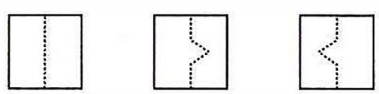
$$X_{k+1} = \begin{pmatrix} X_k & O_k \\ O_k & X_k \end{pmatrix} \text{ and } O_{k+1} = \begin{pmatrix} O_k & X_k \\ X_k & 4O_k \end{pmatrix},$$

for $k = 0, 1, \ldots, m-3$. Here $\|N\|$ denotes the sum of elements of matrix N.

Oh and colleagues go beyond enumeration by bounding the growth rate of knot mosaics [6, 8, 1], and Oh further adapts techniques similar to Section 4 method to solve problems in monomer and dimer tilings [5, 7]. Related ideas were independently used to enumerate the number of rectangular partitions of a rectangle in [10] for the sequence A182275 on the OEIS [4].

Enumerating the number of polygon-free mosaics in certain extensions of the tileset $\mathbb{T}$ is simple to do. For example, if we extend $\mathbb{T}$ to include three types of tiles that connect the top and bottom midpoints like below

Nice!



one simply needs to adjust the v from Definition 5.1 to have $v\left(\begin{smallmatrix} 0 & 1 \\ 0 & 1 \end{smallmatrix}\right) = v\left(\begin{smallmatrix} 1 & 0 \\ 1 & 0 \end{smallmatrix}\right) = 3$.

We point out it is not possible to extend our results naively to the Lomonaco and Kauffman [3] tile set to enumerate the number of *knot-free mosaics*. This is specifically due to the deductions for Case 7e and Case 7f in the proof of Theorem 5.2, as knots allow for additional subcases in which the number of knots does not change.

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colon or period!

7 Appendix

Proposition: The Inclusion-Exclusion Principle For finite sets $A_1, A_2, \ldots, A_M$,

$$\left|\bigcup_{j=1}^{M} A_j\right| = \sum_{\emptyset \neq I \subseteq \{1,2,\ldots,M\}} (-1)^{|I|-1} \left|\bigcap_{j \in I} A_j\right|$$

Proof. The $M=1$ case ~~of one set~~ is trivial, and the case $M=2$ ~~of two sets~~ is the familiar and easily justified statement that $|A_1 \cup A_2| = |A_1| + |A_2| - |A_1 \cap A_2|$. Proceeding inductively, for $M \geq 3$, assume the result for $M-1$ sets, and let us prove it for M sets. Let us split up the sum on the right into those sets I that do not contain M and those that do. By our inductive hypothesis, the first sum equals $|\bigcup_{j=1}^{M-1} A_j|$ and the second sum can be written as $\sum_{\{M\} \subseteq I \subseteq \{1,2,\ldots,M\}} (-1)^{|I|-1} \left|\bigcap_{j \in I} A_j\right|$. Separating out the term with $I = \{M\}$, this equals $|A_M| - \sum_{\emptyset \neq I \subseteq \{1,2,\ldots,M-1\}} (-1)^{|I|} |\bigcap_{j \in I} (A_j \cap A_M)|$. Again using our inductive hypothesis, the sum here is $\left|\bigcup_{j=1}^{M-1} (A_j \cap A_M)\right| = \left|\left(\bigcup_{j=1}^{M-1} A_j\right) \cap A_M\right|$. The entire right side of our desired equation therefore equals $\left|\bigcup_{j=1}^{M-1} A_j\right| + |A_M| - \left|\left(\bigcup_{j=1}^{M-1} A_j\right) \cap A_M\right|$. By the $M=2$ case, ~~the case of our proposition with two sets~~, this equals $|\bigcup_{j=1}^{M} A_j|$, as desired. $\square$

A

Comments on the proof of Prop 4.1

1) Well, the $n=1$ case fails. Perhaps B_1 and C_1 are defined wrong and their definitions should be switched. It's a bad sign that it's not easy to tell what the fix should be.

2) Better to explain to the reader that there are sixteen cases, all similar. Explain what the sixteen cases are and then say "we'll do two for illustration."
As the proof reads now, readers will be very confused by beginning by looking at the $(i, j+2^{n-1})$ entry of A_{n+1}.

3) Does $2^{n-1}+j$ read better than $j+2^{n-1}$? Similar $2^{n-1}+i$ versus $i+2^{n-1}$.